

CE 568P: Experimental Structural Dynamics

Course Code: CE 568P

Course Name: Experimental Structural Dynamics

L-T-P-C: 0-0-4-2

Prerequisites: Basic Structural Dynamics or equivalent

Intended for: M.Tech (Structural Engineering) First-Year Students, Second Semester

Approval: 61st BoA

Preamble

This course adopts an experimental approach to structural dynamics. The suite of experiments are designed to illustrate several aspects of vibratory behaviour of engineering structures subjected to earthquake induced support motions. Students first observe how archetypal structural models respond to controlled vibration or seismic input, then analyze measured data to identify natural frequencies, damping ratios, mode shapes, and other dynamic behaviours. Emphasis is placed on connecting physical observations with idealized analytical models, understanding resonance, damping, and modal interactions directly from experiments. Students learn to operate sensors, data acquisition systems, and analysis tools. This hands-on framework develops deep physical intuition, curiosity-driven learning, and practical insights for applications in earthquake resistant design of building frames, bridges, wave amplification through supporting medium, and water tanks. The experiments are designed to aid an understanding of problems of structural dynamics as relevant to earthquake engineering applications.

Laboratory/Practical/Tutorial Modules

1. Dynamic response of a three-storied building frame under harmonic base motion
2. Behaviour of an asymmetric single-story frame under harmonic excitation
3. Response of a building frame to non-harmonic periodic base motion
4. Study of vibration isolation in secondary systems
5. Dynamic performance of vibration absorber systems
6. Behaviour of frames with soft or weak ground floors
7. Seismic wave amplification and soil–structure interaction
8. Motion of free-standing rigid bodies under base excitations
9. Vibration of beams subjected to harmonic loading
10. Sloshing of liquid storage tanks under dynamic support motions.

Laboratory Infrastructure

- Electromechanical shake tables (horizontal and vertical)
- Accelerometers, signal conditioning amplifiers, and data acquisition systems

- Oscilloscopes and software such as MATLAB or other coding platforms for response analysis
- Model frames, beams, tanks, and isolation systems based on IISc-developed educational modules

Text Books

1. Paz, M., *Structural Dynamics*, CBS Publishers, New Delhi, 1984
2. Chopra, A. K., *Dynamics of Structures: Theory and Applications to Earthquake Engineering*, Pearson Education, India, 2012

References

1. Manohar, C. S., and Venkatesha, S., *Development of Experimental Setups for Earthquake Engineering Education*, Department of Civil Engineering, IISc Bangalore, 2006
2. Clough, R. W., and Penzien, J., *Dynamics of Structures*, Computers and Structures Inc., USA, 2003